

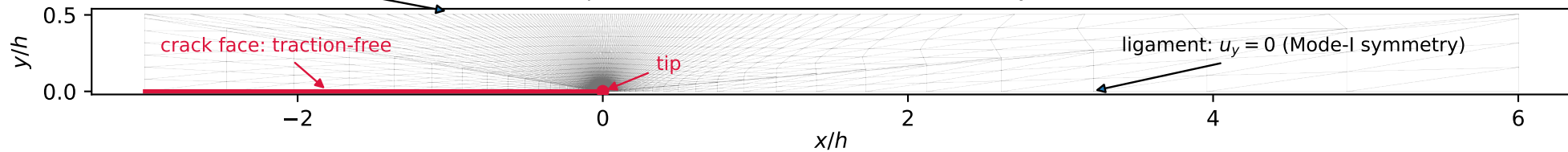
grip: $u_x = 0$, $u_y = \delta$

(a) strip half-model: rosette mesh and boundary conditions

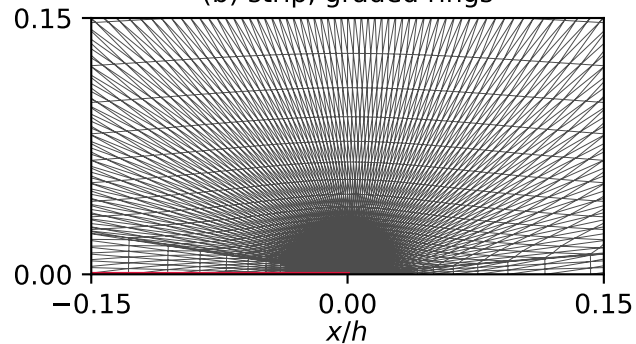
crack face: traction-free

tip

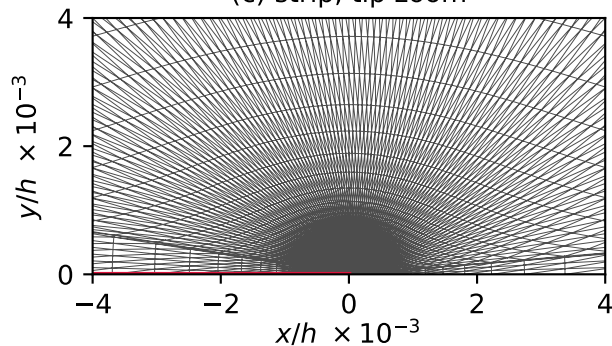
ligament: $u_y = 0$ (Mode-I symmetry)



(b) strip, graded rings



(c) strip, tip zoom



(d) disk, tip zoom

